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Unit of rate of heat transfer = kJ/s or kW

Mode of heat transfer

- ① Conduction [Through ^{material} ~~medium~~ solid]
- ② Convection [Through fluid]
- ③ Radiation

Fourier law of conduction

Heat transfer $\propto$ temp. gradient.

$$Q = -KA \frac{dT}{dx}$$

Q - Conduction heat transfer (W)

K - Thermal conductivity [W/mK] ^{material property}

-ve sign indicates heat flow direction

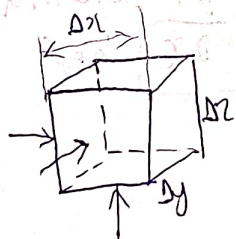
A - Cross sectional area through which heat transfer occurs

Insulators are materials with less thermal conductivity.

Conductivity changes with temperature,

$$K(T) = K_0 (1 + \beta T)$$

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In x direction,
Entering, $Q_x = -KA \frac{\partial T}{\partial x}$

Leaving, $Q_{x+\Delta x} = Q_x + \frac{dQ_x}{dx} \Delta x$

$$Q_{x+\Delta x} = -KA \frac{\partial T}{\partial x} + -KA \frac{\partial^2 T}{\partial x^2} \Delta x$$

any

In y direction

$$Q_{y+\Delta y} = (-KA \frac{\partial T}{\partial y} + -KA \frac{\partial^2 T}{\partial y^2} \Delta y)$$

In z direction

$$Q_{z+\Delta z} = (-KA \frac{\partial T}{\partial z} + -KA \frac{\partial^2 T}{\partial z^2} \Delta z)$$

Heat generation per unit volume = $\dot{q}$

Total heat generation, $Q_{gen} = \dot{q} \Delta x \Delta y \Delta z$.

Mass of elemental volume = $\rho \Delta x \Delta y \Delta z$ ($\rho = \frac{m}{V}$)

Change in internal energy [mcdT] = $\rho \Delta x \Delta y \Delta z \frac{\partial T}{\partial \tau}$
 Energy balance.

$$Q_x + Q_y + Q_z + Q_{gen} = I.E + Q_{x+\Delta x} + Q_{y+\Delta y} + Q_{z+\Delta z}$$

$$-KA \frac{\partial T}{\partial x} - KA \frac{\partial T}{\partial y} - KA \frac{\partial T}{\partial z} + \dot{q} \Delta x \Delta y \Delta z =$$

$$K \frac{\partial^2 T}{\partial x^2} + K \frac{\partial^2 T}{\partial y^2} + K \frac{\partial^2 T}{\partial z^2} + \dot{q} = \frac{\partial T}{\partial \tau} \rho C$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{K} = \frac{\rho C}{K} \frac{\partial T}{\partial \tau}$$

Assume, conductivity, ρ are constant.

$$\frac{K}{\rho C} = \alpha \quad \text{Thermal diffusivity}$$

Cartesian system

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{K} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$$

General heat conduction in Cartesian.

If no heat generation,

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$$

For a steady state conduction,

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0 \quad \left[\frac{\partial T}{\partial \tau} = 0 \right]$$

For steady state, no heat generation in 2D,

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

In 1D, $\frac{\partial^2 T}{\partial x^2} = 0$

General heat conduction in polar coordinates [cylindrical coordinates]

Cylindrical coordinates T is $T(r, \theta, z)$.

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Spherical coordinates

$$T = T(r, \theta, \phi)$$

General heat conduction eqⁿ,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2 T}{\partial \phi^2} = 0$$

Boundary conditions

① Constant temperature $T = T_1$

② Constant heat flux $q_r = \frac{Q}{A} \text{ (W/m}^2\text{)}$

③ Convective heat transfer, $Q = h A_s (T - T_\infty)$

④ Radiative heat transfer.

⑤ Combination of 1, 2, 3, 4

One dimensional steady heat transfer

① One dimensional temp. $T = T(x)$

② $\frac{\partial^2 T}{\partial x^2} = 0$

③ $\frac{dT}{dx} = C_1$

$$④ T = C_1 x + C_2$$

⑤ C_1, C_2 depends of boundary conditions.

⑥ For boundary condition,

$$x=0, T=T_1 \text{ and } x=L, T=T_2.$$

$$T_1 = C_2$$

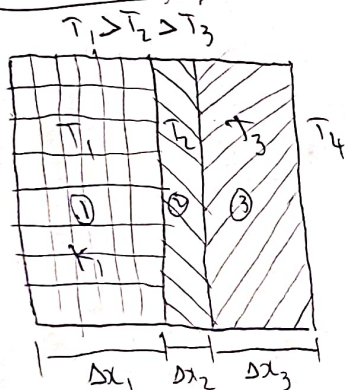
$$T_2 = C_1 L + T_1$$

$$T = \frac{(T_2 - T_1)x}{L} + T_1$$

$$\text{Heat transfer, } Q = \frac{KA(T_1 - T_2)}{L}$$

$$\left[\begin{array}{l} \text{In table,} \\ Q = \frac{\Delta T}{R} \text{ where } R = \frac{L}{KA} \end{array} \right] \quad \text{Thermal resistance}$$

Composite walls



① Parallel plates

② Same heat flow.

$$Q = \frac{K_1 A (T_1 - T_2)}{\Delta x_1} \Rightarrow T_1 - T_2 = \frac{Q \Delta x_1}{K_1 A} \quad \text{--- (1)}$$

$$Q = \frac{K_2 A (T_2 - T_3)}{\Delta x_2} \Rightarrow T_2 - T_3 = \frac{Q \Delta x_2}{K_2 A} \quad \text{--- (2)}$$

$$Q = \frac{K_3 A (T_3 - T_4)}{\Delta x_3} \Rightarrow T_3 - T_4 = \frac{Q \Delta x_3}{K_3 A} \quad \text{--- (3)}$$

$$\text{① + ② + ③} \Rightarrow T_1 - T_4 = \frac{Q}{A} \left(\frac{\Delta x_1}{K_1} + \frac{\Delta x_2}{K_2} + \frac{\Delta x_3}{K_3} \right)$$

$$\Rightarrow Q = \frac{T_1 - T_4}{\frac{\Delta x_1}{K_1 A} + \frac{\Delta x_2}{K_2 A} + \frac{\Delta x_3}{K_3 A}}$$

$$Q = \frac{\Delta T_{\text{Overall}}}{\sum R_{\text{Overall}}}$$



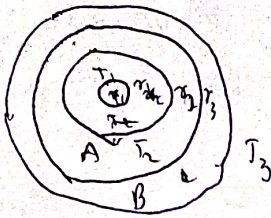
1D steady heat transfer - cylinders



Radial heat transfer.

$$Q_r = \frac{2\pi KL(T_i - T_o)}{\ln(r_o/r_i)}$$

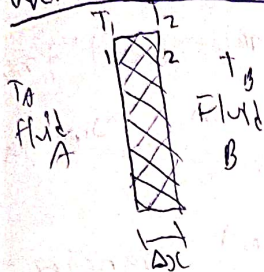
Composite cylinder, heat transfer,



$$Q_r = \frac{2\pi L(T_1 - T_3)}{\frac{\ln(r_2/r_1)}{k_1 A_1} + \frac{\ln(r_3/r_2)}{k_2 A_2}}$$

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Convection heat transfer
Overall heat transfer



Newton's law of cooling

$$Q = hA(T_w - T_\infty)$$

h - Convective heat transfer coe. (W/m²K)

$$Q_{\text{con}} = h_A A (T_1 - T_1)$$

$$Q_{\text{con}} = K A (T_1 - T_2)$$

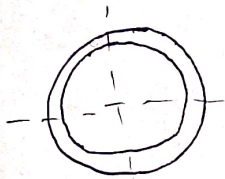
$$Q_{\text{con}} = h_B A (T_2 - T_2)$$

$$Q = \frac{T_A - T_B}{\frac{1}{h_A A} + \frac{\Delta x}{KA} + \frac{1}{h_B A}}$$

$$\Rightarrow Q = U A \Delta T_{\text{overall}}$$

U ⇒ Overall heat transfer coe.

$$U = \frac{1}{\frac{1}{h_A} + \frac{\Delta x}{K} + \frac{1}{h_B}}$$



$$Q = h_i A_i (T_i - T_i)$$

$$Q = \frac{2\pi K L (T_1 - T_2)}{\ln(r_o/r_i)}$$

$$Q = h_o A_o (T_2 - T_o)$$

For cylinders,

$$\Rightarrow Q = \frac{T_i - T_o}{\frac{1}{h_i A_i} + \frac{\ln(r_o/r_i)}{2\pi K L} + \frac{1}{h_o A_o}}$$

$$\times \frac{A_o}{A_o} \Rightarrow Q = \frac{A_o (T_i - T_o)}{\frac{A_o}{h_i A_i} + \frac{A_o \ln(r_o/r_i)}{2\pi K L} + \frac{1}{h_o}}$$

$$\Rightarrow U_o = \frac{1}{\frac{A_o}{h_i A_i} + \frac{A_o \ln(r_o/r_i)}{2\pi K L} + \frac{1}{h_o}}$$

$$Q = U_o A_o (T_i - T_o)$$

$U_o \rightarrow$ overall heat transfer co.
based on outside surface

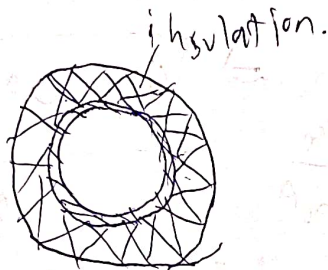
$$\textcircled{1} \times \frac{A_i}{A_i} \Rightarrow Q = A_i (T_i - T_o)$$

$$\frac{A_o}{A_i} \frac{1}{h_i} + \frac{A_i \ln(r_o/r_i)}{2\pi k L} + \frac{A_i}{A_o h_o}$$

$$Q = U_i A_i (T_i - T_o)$$

$U_i \rightarrow$ overall heat transfer co.
based on inside surface

Critical radius of insulation [max. heat transfer occurs]

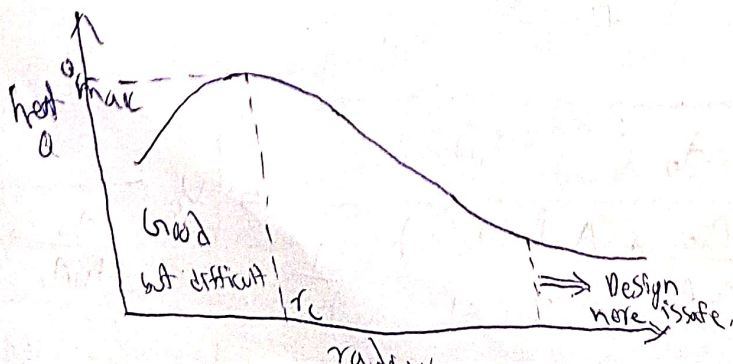


$$Q = \frac{2\pi L (T_i - T_o)}{\left(\frac{\ln(r_o/r_i)}{k} + \frac{1}{r_o h} \right)}$$

$$\frac{dQ}{dr_o} = 0 \Rightarrow$$

$$\frac{dQ}{dr_o} = \frac{-2\pi L (T_i - T_o) \left(\frac{1}{kr_o} - \frac{1}{hr_o^2} \right)}{\left[\frac{\ln(r_o/r_i)}{k} + \frac{1}{r_o h} \right]^2}$$

$$\Rightarrow \frac{1}{kr_o} - \frac{1}{hr_o^2} = 0$$



Critical radius is applied on electric conductors.

Assignment March 3 ~~March 3~~ ① Write briefly about different types of heat exchangers with sketches.

② Take the picture of an automobile radiator explain its parts, identify the heat transfer elements, how the temp. of coolant is controlled.

③ Take the picture of a split air conditioner condenser, identify the heat transfer elements.

④ Take the picture of a solar collector and explain its working and heat transfer mechanism.

⑤ Take the picture of a cooling tower and explain its working.

Problems

10 One face of a copper plate $10\text{cm} \times 25\text{cm} \times 3\text{cm}$ thick is maintained at 400°C . The other face is maintained at 100°C . Calculate the heat transfer rate through the plate.

$$T_1 = 673\text{K}$$

$$T_2 = 373\text{K}$$

$$A = 0.1 \times 0.25 = 0.025\text{m}^2$$

Assuming from data book $K = 386\text{W/mK}$

$$Q = \frac{-KA(T_2 - T_1)}{\Delta x} = \frac{-386 \times 0.025 \times (373 - 673)}{0.03} = 96\text{KW}$$

① An exterior of a house may be approximated by a 10.2cm layer of common brick ($K = 0.7\text{W/mK}$) followed by a 3.8cm layer of gypsum ($K = 0.48\text{W/mK}$). What thickness of loosely packed rock wool insulation ($K = 0.065\text{W/mK}$) should be added to reduce the heat loss/gain through the wall

by 80%? $\frac{Q_{\text{with insul}}}{Q_{\text{without insul}}} = 0.2$

(0.08m)

